TEXORA · LECTURE SHEET

Introduction to Knitting Technology

Knitting builds fabric not by crossing two sets of yarn at right angles, but by drawing one yarn — or a sheet of warp yarns — into loops and intermeshing each new loop through the loops of the previous row. That single difference explains knitting's stretch, softness, low capital route to shaped garments, and its dominant place in apparel.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 1 · LEVEL 3 TERM 1

1. What knitting is

Knitting is the formation of fabric by interlooping: a yarn is bent into a loop, and that loop is drawn through a loop already formed, so the fabric is held together by loops passing through loops. A horizontal row of loops formed in one working cycle is a **course**; a vertical column of loops is a **wale**. Fabric grows course by course, wale by wale. Courses per centimetre (CPC) and wales per centimetre (WPC) together describe how close the structure is; their product is the **stitch density** (loops per cm²).

Because the yarn in each loop is curved, it can straighten under tension: knitted fabric therefore extends readily, recovers well when the loops are not distorted past their elastic range, drapes softly, traps still air in the loop spaces (warmth), and rarely creases sharply. The same curvature is the weakness — a broken loop can ladder (run) down a wale, and an unsupported edge curls.

2. Knitting versus weaving

FEATURE	KNITTING	WEAVING
Yarn systems	One system (weft knitting) or one warp sheet guided loop-by-loop (warp knitting)	Two systems: warp and weft, interlaced at right angles
Unit of structure	Loop intermeshed through the previous loop	Interlacement (plain, twill, satin) of straight yarns
Extensibility	High, especially widthwise in weft knits	Low except on the bias
Yarn preparation	Minimal — yarn from cone/package can be knitted directly (weft)	Warping and sizing normally required
Shaping	Fully-fashioned shaping and whole garments possible	Fabric is flat; shaping only by cutting and sewing

Production character	High output per machine on circular frames; easy pattern change	High output; pattern change slower
Typical faults	Laddering, curling, spirality, barré	Warp/weft defects, reed marks

3. The two families: weft and warp knitting

Weft knitting: one yarn (or one yarn per feeder) travels across the machine and is formed into a whole course of loops in sequence or simultaneously across the needle bed. Courses build one on the next. Plain jersey, rib, interlock, purl and all flat-bed and hosiery work are weft knitting. A weft knit can, in principle, be unravelled from the last course.

Warp knitting: every needle has its own yarn, supplied from a warp beam. All loops of a course are formed at the same instant as guide bars swing the yarns to the needles and then shift (shog) to connect one wale to the next. Tricot and raschel are the warp families. Warp knits are far more run-resistant and dimensionally stable, and are produced at very high course rates, but pattern and yarn preparation are more committed.

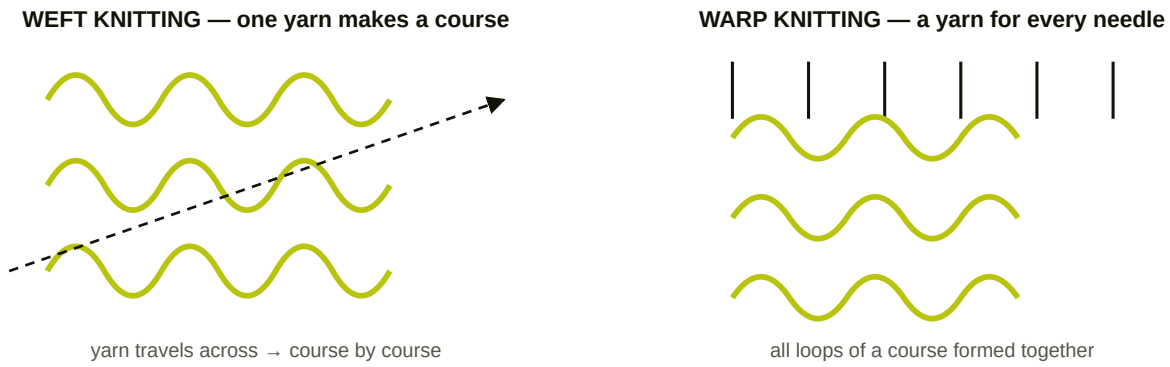


Figure 1 — The two families. Weft: a single yarn is looped across the width. Warp: a sheet of parallel yarns, one per needle, is looped simultaneously and connected sideways by guide-bar shifts.

4. The loop — parts and language

One knitted loop (stitch) consists of a **needle loop** (the head and two limbs, formed in the needle hook) and a **sinker loop** (the yarn connecting the foot of one needle loop to the next). The length of yarn in one complete loop is the **stitch length (loop length)**, l — the single most important quality variable in knitting: it fixes courses and wales per cm after relaxation, fabric weight, width, cover and much of dimensional behaviour. Shorter l → tighter, heavier, narrower, more stable fabric; longer l → looser, lighter, wider, softer fabric. Stitch length is set by the stitch cam (weft) or by run-in (warp) and is measured as yarn length per loop, in mm.

$$\text{Stitch density} = \text{CPC} \times \text{WPC} \ \ (\text{loops/cm}^2) \quad \cdot \quad \text{Tightness factor K} = \sqrt{\text{tex}} \div l\ (\text{mm})$$

Worked check. A jersey shows CPC = 15/cm and WPC = 12/cm, so stitch density = 15 × 12 = **180 loops/cm²**. With 19.7 tex yarn at l = 4.2 mm, K = √19.7 ÷ 4.2 = 4.44 ÷ 4.2 ≈ **1.06** — a medium-tight plain jersey (typical jersey K ≈ 0.9–1.2 in these units).

5. Needles — the heart of the machine

- **Bearded (spring-beard) needle:** thin hook whose beard is pressed closed by a presser to trap the yarn. Historically first; still used in some warp and special machines. Fine gauges possible; loop formation needs separate presser and sinker elements.
- **Latch needle:** hook closed by a pivoted latch that the old loop itself opens and closes. Self-acting — it made modern high-speed weft and warp knitting possible. The standard needle of this course (full cycle in Topic 3).
- **Compound needle:** hook closed by a sliding closing element moving inside a grooved stem. Short, controlled stroke, very high speeds; used on modern tricot warp machines and some circular machines.

Needles work with **sinkers** (hold the fabric down and form the sinker loop), **cams** (drive needle butts through clearing, loop-forming and knock-over positions), jacks/selectors (pattern selection) and, in warp knitting, **guide bars** that lay the yarn. **Machine gauge** is the number of needles per unit length — per inch (E) on most machines: E28 means 28 needles per inch, pitch = 25.4 ÷ 28 = 0.907 mm. Finer gauge needs finer yarn: a working rule matches yarn count to gauge so the yarn fills the hook without crowding (Topic 12 uses gauge in every production calculation).

6. Where knitting is used — and why it grew

Knitting dominates T-shirts, polos, underwear, socks and hosiery, sportswear, sweaters and an increasing share of dresses, leggings and seamless whole garments; warp knits serve lingerie, nets, lace, swimwear, automotive and technical uses. Drivers: comfort stretch without elastane (and far more with it), direct cone-to-fabric production with almost no yarn preparation in weft knitting, fast style change, and machines that knit shaped panels or complete garments, cutting cut-and-sew waste. Bangladesh's knit sector — built largely on circular single-jersey, rib and interlock machines plus flat-bed collars and cuffs — is the industrial context for every topic that follows.

Glossary

Course — horizontal row of loops. **Wale** — vertical column of loops. **CPC / WPC** — courses / wales per cm. **Stitch (loop) length, l** — yarn length in one complete loop, mm. **Needle loop** — head and

limbs of the loop formed in the hook. **Sinker loop** — yarn joining the feet of adjacent needle loops. **Gauge (E)** — needles per inch. **Tightness factor** — √tex ÷ l; a compactness index. **Laddering** — a broken loop running down its wale. **Fully-fashioned** — panel shaped on the machine by adding/removing wales.

Key points

• Knitting = interlooping; fabric is described by courses, wales, stitch density and stitch length.
• Stitch length is the master variable — it sets weight, dimensions, cover and stability after relaxation.
• Weft knitting loops one yarn across; warp knitting loops a whole yarn sheet at once and is run-resistant.
• Latch, bearded and compound needles are the three needle principles; gauge links needle spacing to suitable yarn count.
• Stretch, softness, shaping and low yarn preparation explain knitting's apparel dominance — paid for with laddering, curling and spirality risks.

Practice

1. A fabric has CPC 18/cm and WPC 14/cm. Find its stitch density. If stitch length rises 10 % at the same yarn, state the direction of change in CPC, WPC and fabric weight, with reasons.
2. Calculate the needle pitch of E24 and E32 machines in mm, and state which must use the finer yarn.
3. Explain, from loop geometry, why a weft knit stretches widthwise more readily than a woven fabric of the same yarn.
4. Give two products that must be warp knitted rather than weft knitted, with the property that decides each.
5. A buyer complains a jersey is "too loose and grows in wear". Which single machine setting is checked first, and which way should it move?

Answer key

1. 18 × 14 = **252 loops/cm²**. Longer loops are larger loops: both CPC and WPC fall (fewer, bigger loops per cm), and weight per m² falls because the same yarn length is spread over a larger area.
2. Pitch = 25.4 ÷ E. E24: 25.4 ÷ 24 = **1.06 mm**. E32: 25.4 ÷ 32 = **0.79 mm**. E32 has the smaller hook and pitch, so E32 takes the finer (lower tex / higher Ne) yarn.
3. Each loop stores yarn in a curved configuration. Widthwise tension straightens the limbs and pulls yarn from the loop head, so the wales separate with little yarn extension. In a weave the weft is already nearly straight, so widthwise extension requires yarn elongation itself — much smaller.

4. Examples: fine lingerie/net and lace (run-resistance and stability of tricot/raschel — a weft knit would ladder); swimwear and elastic nets (stable, non-curling warp structure); car seat and technical meshes (dimensional stability). Any two with the stability/run-resistance reason earn the point.
5. Stitch length (stitch cam setting). It is almost certainly too long; shorten l to raise CPC/WPC, tightness factor and stability — then re-check weight and width, which will rise and fall respectively.

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